MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG3DSESTA200 – Statistical techniques for Data Science and Machine Learning (Data analytics Specialization)

Duration: 1Hour Maximum: 30 Marks

PRACTICAL EXAMINATION

Answer any five questions. Each question carries 6 marks.

Solve the questions using Python in the computer lab. The data in the question, along with their answer, must be written in the answer paper

1. Consider the salary dataset build a multiple regression model to predict the salary of an individual based on their age and years of experience. [A][5]

2. Survey of 320 families with 5 children each revealed the following distribution

No of boys	5	4	3	2	1	0
No of girls	0	1	2	3	4	5
No of families	14	56	110	88	40	12

   Is the result consistent with the hypothesis that male and female births are equally probable? [A][5]

3. Test whether the average height of adults in the city significantly different from the national average height of 175 cm by collecting a sample of 100 adults from the city and measure their heights. Assume the sample mean height is 180 cm with a standard deviation of 8 cm. [A][5]

4. Test whether there is a significant correlation between the no. of hours studied and exam scores, [A][5]

Hours Studied	2	4	6	3	5	1	7	8	4	6
Exam score	60	80	90	70	85	50	95	98	75	92

5. Ten students are selected at random from a school and their heights are found to be in inches:
   51, 52,52,54,55,56,57,58,58, and 59.
   In light of these data, discuss the suggestion that the mean height of the students of the school is 55inches. You may use 5% level of significance. Write the Python program used for testing the hypothesis and give the output obtained. [A][5]

6. The time taken by workers in performing a job by method 1 and method 2 are given below.

Method 1	20	16	26	25	23	
Method 2	28	33	42	35	52	34

   Does the data show that the variance of time distribution by the two methods does not differ significantly? [A][5]

7. Given the following data relating to social status and state of intelligence. Test whether intelligence is related to social status. [A][5]

	Dull	Average	Brilliant
Lower Middle	22	35	23
Middle	38	70	32
Upper Middle	60	20	20

8. The following table shows the lives (in hours) of four batches of electric lamps:
   BATCH 1:1600 1610 1650 1680 1700 1720 1800
   BATCH II :1580 1640 1640 1700 1750
   BATCH III:1460 1550 1600 1620 1640 1660 1740 1820
   BATCH IV: 1510 1520 1530 1570 1600 1680

   Perform an analysis of variance of these data and show that a significant test does not reject their homogeneity.
   Write the Python program used for the analysis and the output obtained. [A][5]